

CBSC-ZDC v2.2: An Audited, CLI-Ready Specification for a Four-Momentum-Conditioned Fast Monte Carlo Surrogate of Neutron Showers in a Zero Degree Calorimeter

Audited experimental specification

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Abstract

We specify a falsifiable experiment for learning the scored detector-readout distribution of non-pencil-beam single-neutron Geant4 events in a fixed Zero Degree Calorimeter (ZDC). The target detector has 65 longitudinal layers and 6,790 valid readout channels (400 ECAL and 6,390 HCAL). The available corpus is expected to contain approximately 765,000 events over 0–300 GeV incident neutron kinetic energy; final claims are restricted to 50–250 GeV. The sole raw event condition is the neutron four-momentum. This revision reconciles four independent audits line by line and removes six blocking defects in the previous specification: mixed raw/thresholded energy accounting, an unidentified shared latent, double-stochastic support generation, absent numerical decision gates, kinetic/total-energy conflation, and an unobserved reserve variable. The resulting hierarchy generates visible response, total response, first positive layer, active layers, arbitrary nonnegative longitudinal budgets, exact finite hit counts, exact support, and positive energy shares. Deposited energy is not forced to decrease with depth; only the remaining accounting budget is non-increasing. Binary support is produced once by a supervised geometry-aware scorer and Gumbel-Top- k sampling, while conditional flow matching is restricted to continuous positive-energy shares. The repository implements the complete CLI path from ROOT inspection through geometry freezing, conversion, leakage-aware splitting, data audit, provenance freezing, training, sampling, evaluation, and timing. Algebraic invariants and synthetic software execution are verified, but no Geant4-fidelity or speed claim is made before production training and frozen held-out evaluation.

Contents

1	Status, scope, and exact claim	3
1.1	Research question	3
1.2	Claims that are presently supported	3
2	Audit reconciliation and design corrections	3
3	Frozen data and detector contract	4
3.1	Incident energy convention	4
3.2	ROOT and target contract	4
3.3	Geometry	4
3.4	Split and leakage control	4
4	Conditional factorization without an unidentified latent	5
5	Longitudinal cascade	5
5.1	Visibility and total response	5
5.2	First positive layer and activity	5
5.3	Arbitrary nonnegative profile	5

6	Count, support, and continuous share generation	6
6.1	Finite count model	6
6.2	One stochastic support mechanism	6
6.3	Conditional flow matching for positive shares	6
7	Exact-support decoder and anti-dust theorem	6
7.1	Raw-deposit mode	6
7.2	Thresholded-readout-only mode	6
8	Training objectives and exposure control	7
9	CLI and production procedure	7
10	Evaluation and predeclared decisions	8
10.1	Structural gates	8
10.2	Distributional evaluation	8
10.3	Primary gates	8
11	Baselines and external context	8
12	Software QA and verified execution	8
13	Residual risks and falsification paths	9
14	Auditor checklist	9
15	Conclusion	9
A	Tensor and artifact contract	10
B	Experiment matrix	10
C	Failure-mode matrix	11
D	Audit-to-code traceability	11
E	Production command transcript template	11
F	Evidence boundary for the supplied fixture	12

1 Status, scope, and exact claim

1.1 Research question

Let $Y \in \mathbb{R}_{\geq 0}^{6790}$ denote the frozen detector readout, $p^\mu = (E_{\text{tot}}, p_x, p_y, p_z)$ the neutron four-momentum, and $\mathcal{G}$ the immutable detector geometry. The empirical question is whether a learned sampler can approximate

$$p_\theta(Y \mid p^\mu, \mathcal{G}) \approx p_{\text{G4}}(Y \mid p^\mu, \mathcal{G}) \quad (1)$$

over $50 \leq K_{\text{inc}} \leq 250$ GeV, while preserving exact structural contracts and providing a useful end-to-end speedup.

This document and repository are an implementation-ready experimental scaffold, not a trained simulator. “Implementation-ready” means that the remaining software operation is to install the declared dependencies, execute the CLI against the production corpus, and run the frozen training/evaluation plan. It does not mean that the production ROOT contract, detector geometry, fidelity, or latency has already passed. The operational source of truth for agents is `docs/IMPLEMENTATION_GUIDE.md`, supported by `AGENTS.md`, the data contract, loss-weight protocol, evaluation protocol, Vertex runbook, and troubleshooting guide.

1.2 Claims that are presently supported

1. The event-level longitudinal deposits D_ℓ must not be constrained by $D_{\ell+1} \leq D_\ell$ for neutron-induced hadronic showers.
2. The revised decoder gives algebraically exact finite support and budget closure under its feasibility conditions; implementation checks use a declared floating-point tolerance.
3. The active code has a complete CLI, deterministic provenance, unit tests, and an end-to-end synthetic smoke path.
4. The experiment is falsifiable through predeclared absolute and comparative gates.

No current artifact supports a claim of Geant4 fidelity, downstream reconstruction equivalence, absence of memorization, or speedup.

2 Audit reconciliation and design corrections

Four supplied audits contained contradictory overall verdicts but overlapping technical findings. Each was read in full and adjudicated at finding level. The strongest audit identified six blockers, all accepted after independent re-derivation. Their disposition is summarized in [table 1](#).

Table 1: Blocking defects and v2.2 corrections.

ID	Defect	Correction
B1	Raw layer energy, thresholded cell output, and a subthreshold residual were mixed inconsistently.	Raw deposit is primary; thresholded-readout-only is a separate target; no mixed residual.
B2	A shared latent z had neither marginal likelihood nor inference mechanism and could be ignored.	z is removed; dependence is explicit through ancestral conditioning.
B3	Binary support was a singular continuous-flow target and was perturbed again by independent Gumbel noise.	A supervised support scorer plus one Gumbel-Top- k draw generates support; flow models shares only.
B4	No numerical success, non-inferiority, or failure gates were declared.	Primary gates are frozen before final-test exposure.
B5	Geant4 kinetic energy was conflated with the total-energy component of a four-vector.	$p_0^\mu = E_{\text{tot}}$; range selection/reporting use $K_{\text{inc}} = E_{\text{tot}} - m_n$.
B6	A softmax reserve was positive, unobserved, and non-identifiable.	Reserve is removed; generated layer budgets and cell outputs close directly to total response.

Additional accepted corrections include distinct notation for the first positive layer and support set, dimensionless share transforms, explicit treatment of the zero residual-share case, bidirectional layer context as

the primary model, dynamic count feasibility masks, required point-cloud comparison, independent seed repetitions, immutable final-test metrics, and corrected bibliography metadata. Original material is retained only under a clearly marked legacy directory.

3 Frozen data and detector contract

3.1 Incident energy convention

Geant4 particle-gun energy is conventionally kinetic energy, while a four-vector stores total energy. We therefore define

$$p^\mu = (E_{\text{tot}}, p_x, p_y, p_z), \quad E_{\text{tot}}^2 - \|\mathbf{p}\|^2 = m_n^2, \quad K_{\text{inc}} = E_{\text{tot}} - m_n, \quad (2)$$

with $m_n = 0.93956542052 \text{ GeV}$. The mass shell is accepted by a float64 comparison of E_{tot} with $\sqrt{\|\mathbf{p}\|^2 + m_n^2}$, avoiding ill-conditioned float32 subtraction at high energy.

The available support and primary domain always refer to K_{inc} . The model condition remains the full p^μ . Derived event features must be deterministic functions of p^μ only.

3.2 ROOT and target contract

The default adapter targets PODIO/EDM4hep-style branches for one generator neutron, event headers, and ECAL/HCAL `SimCalorimeterHit` collections. The stored hit energy and position are expected in GeV and mm, respectively, consistent with the EDM4hep data model, but production files must be inspected rather than assumed [1].

The primary target is raw deposit:

$$Y_i = \sum_{h: c(h)=i} E_h \mathbf{1}[E_h > 0], \quad T = \sum_{i=1}^{6790} Y_i. \quad (3)$$

Repeated channel IDs are summed. Zero-response events are retained. A separate thresholded-readout-only experiment may define $Y_i^{(\tau)} = Y_i \mathbf{1}[Y_i \geq \tau]$ and $T_\tau = \sum_i Y_i^{(\tau)}$, but it may not allocate raw energy into thresholded cells or introduce an unobserved residual.

3.3 Geometry

The strict geometry contains 65 layers with channel counts

$$(N_0, \dots, N_{64}) = (400, \underbrace{100, \dots, 100}_{63 \text{ layers}}, 90), \quad (4)$$

so $\sum_\ell N_\ell = 6790$. Channel identity is the pair (collection, cellID). A signed sentinel -100 and its uint64 representation are excluded before conversion.

Median observed positions freeze the channel map. Physical lateral k -nearest-neighbor edges and adjacent-layer nearest-neighbor edges are deterministic functions of the frozen positions. Arrays, edges, and features are covered by a geometry SHA-256 hash. Any change defines a new experiment.

3.4 Split and leakage control

The preferred split unit is an independent Geant4 run/job/seed family represented by source file and run number. Deterministic event hashing is an explicitly disclosed fallback. The final test split is not used for target choice, cap estimation, normalization, architecture, objective, solver, checkpoint selection, or gate setting.

4 Conditional factorization without an unidentified latent

The v2.2 generation order is

$$p_\theta(V, T, L_0, A, D, K, H, W, Y \mid p^\mu, \mathcal{G}) \quad (5)$$

$$\begin{aligned} &= p_\theta(V \mid p^\mu) p_\theta(T \mid V, p^\mu) p_\theta(L_0 \mid T, V, p^\mu) p_\theta(A \mid L_0, T, V, p^\mu) \\ &\quad \times p_\theta(D \mid A, T, p^\mu) p_\theta(K \mid D, A, p^\mu) p_\theta(H \mid K, D, A, p^\mu, \mathcal{G}) \\ &\quad \times p_\theta(W \mid H, K, D, p^\mu, \mathcal{G}) \delta(Y - \text{Decode}(W, H, K, D)). \end{aligned} \quad (6)$$

Here V is visibility, T total response, L_0 first positive layer, A active layers, D layer budgets, K counts, H selected support, and W positive shares. There is no hidden common cause whose trainability is left unspecified. Stochastic dependence is induced by the explicit conditional chain.

5 Longitudinal cascade

5.1 Visibility and total response

A Bernoulli hurdle models V . Given $V = 1$, a finite beta mixture models a unit variable $r \in (0, 1)$. We define

$$T = K_{\text{inc}} \rho_{\text{max}} r, \quad (7)$$

where $\rho_{\text{max}} > 0$ is frozen from the complete training split and is not assumed to be one. If production data contain endpoint atoms or tails poorly represented by the beta mixture, those are explicit model failures or motivate a predeclared response ablation.

5.2 First positive layer and activity

L_0 is modeled directly as a categorical layer index, conditional on a visible response. Activity satisfies

$$A_\ell = 0 \text{ for } \ell < L_0, \quad A_{L_0} = 1, \quad (8)$$

while gaps are allowed for $\ell > L_0$. This replaces a renormalized in-detector hazard whose relation to no-response mass was ambiguous.

5.3 Arbitrary nonnegative profile

A conditional flow-matching field generates active-layer logits. A masked simplex yields

$$D_\ell \geq 0, \quad D_\ell = 0 \text{ if } A_\ell = 0, \quad \sum_{\ell=0}^{64} D_\ell = T. \quad (9)$$

No order relation is imposed among the D_ℓ . This is necessary because neutron hadronic interaction depth and final state are stochastic, and event-level deposits can start late, rise, fluctuate, and contain multiple maxima [2, 3, 4].

Proposition 1 (remaining-budget monotonicity). Define $R_\ell = T - \sum_{j=0}^{\ell} D_j$. Then $R_{\ell+1} \leq R_\ell$ and $R_\ell \geq 0$.

Proof. $R_{\ell+1} = R_\ell - D_{\ell+1} \leq R_\ell$ because $D_{\ell+1} \geq 0$. Also, by equation (9), $R_\ell = \sum_{j>\ell} D_j \geq 0$. No relation between $D_{\ell+1}$ and D_ℓ follows. $\square$

6 Count, support, and continuous share generation

6.1 Finite count model

For layer ℓ , K_ℓ is categorical on $\{0, \dots, N_\ell\}$. Before sampling, logits are masked by geometry and budget:

$$A_\ell = 0 \text{ or } D_\ell = 0 \Rightarrow K_\ell = 0, \quad (10)$$

$$\text{raw mode, } D_\ell > 0 \Rightarrow 1 \leq K_\ell \leq N_\ell, \quad (11)$$

$$\text{thresholded-only mode} \Rightarrow K_\ell \leq \lfloor D_\ell / \tau \rfloor. \quad (12)$$

Categorical counts avoid a forced Poisson mean-variance relation. Count calibration is a primary observable, not an internal nuisance.

6.2 One stochastic support mechanism

A graph network produces support logits a_i using static geometry, condition, layer budgets, and count fractions. For each layer, exactly K_ℓ valid cells are drawn without replacement by Gumbel-Top- k [5]. The flow no longer models a binary endpoint, and no second independent Gumbel perturbation is applied.

6.3 Conditional flow matching for positive shares

For selected cells only, a continuous target encodes normalized positive shares. With reference noise X_0 and target X_1 , the straight path is

$$X_t = (1 - t)X_0 + tX_1, \quad u_t = X_1 - X_0, \quad (13)$$

and the field minimizes

$$\mathcal{L}_{\text{FM}} = \mathbb{E} [\|M \odot (v_\theta(X_t, t, c, \mathcal{G}, D, K, H) - u_t)\|_2^2], \quad (14)$$

where M masks non-identifiable entries. The network consumes t explicitly. Flow matching is a simulation-free method for learning continuous normalizing-flow vector fields [6].

The primary backbone uses edge-conditioned message passing and bidirectional joint layer mixing. Causal shallow-to-deep context is retained only as an ablation: physical depth ordering does not require a parallel readout generator to prohibit deep-to-shallow internal computation.

7 Exact-support decoder and anti-dust theorem

Let H_ℓ be a selected set of cardinality K_ℓ , and let

$$w_i = \frac{\exp r_i}{\sum_{j \in H_\ell} \exp r_j}, \quad i \in H_\ell. \quad (15)$$

7.1 Raw-deposit mode

$$Y_i = \begin{cases} D_\ell w_i, & i \in H_\ell, \\ 0, & i \notin H_\ell. \end{cases} \quad (16)$$

7.2 Thresholded-readout-only mode

For $D_\ell \geq K_\ell \tau$,

$$Y_i = \begin{cases} \tau + (D_\ell - K_\ell \tau) w_i, & i \in H_\ell, \\ 0, & i \notin H_\ell. \end{cases} \quad (17)$$

Theorem 2 (support, threshold, and budget). Under the feasibility conditions, equations (16) and (17) produce zero outside H_ℓ , exactly K_ℓ selected entries, nonnegative energies, and $\sum_{i: \ell(i)=\ell} Y_i = D_\ell$. In thresholded-only mode every selected energy is at least τ .

Proof. Unselected entries are defined as zero. Since softmax weights are positive and sum to one, the raw sum is D_ℓ . In thresholded mode,

$$\sum_{i \in H_\ell} Y_i = K_\ell \tau + (D_\ell - K_\ell \tau) \sum_{i \in H_\ell} w_i = D_\ell.$$

Feasibility makes the residual coefficient nonnegative, so selected entries are at least τ . $\square$

This theorem prevents dense “dust” conditional on K_ℓ . It does not make a wrong generated count correct. Counts, support, and positive-hit spectra are therefore trained and reported separately. In floating-point execution, exactness is tested against the declared tolerance.

8 Training objectives and exposure control

The component losses are

$$\mathcal{L} = \lambda_V \mathcal{L}_V + \lambda_T \mathcal{L}_T + \lambda_{L_0} \mathcal{L}_{L_0} + \lambda_A \mathcal{L}_A + \lambda_D \mathcal{L}_D + \lambda_K \mathcal{L}_K + \lambda_H \mathcal{L}_H + \lambda_W \mathcal{L}_W. \quad (18)$$

They correspond to visibility BCE, response likelihood, first-layer CE, activity BCE, profile flow matching, count CE, support BCE/ranking, and share flow matching. Loss weights are explicit in a hashed configuration.

Recommended order:

1. data/geometry/target freeze;
2. response model;
3. longitudinal profile;
4. count model;
5. support scorer;
6. positive-share flow;
7. generated-condition exposure and joint fine-tuning;
8. matched baselines and frozen held-out evaluation.

A decreasing aggregate loss is insufficient: each stage must be reported in free-running mode.

The CLI training implementation includes AdamW, cosine or constant scheduling, mixed precision, gradient accumulation, clipping, early stopping, resume checks, environment capture, and checkpoint/config/geometry hashes. At least three predeclared training seeds are required; the best seed may not be selected after comparison.

9 CLI and production procedure

The repository exposes a native command-line interface:

```
cbssc-zdc inspect-root ...
cbssc-zdc scan-geometry ...
cbssc-zdc convert ...
cbssc-zdc split ...
cbssc-zdc audit-dataset ...
cbssc-zdc freeze-config ...
cbssc-zdc train ...
cbssc-zdc qa ...
cbssc-zdc sample ...
cbssc-zdc evaluate ...
cbssc-zdc benchmark ...
```

The production sequence is strict. ROOT inspection resolves every branch. Geometry freezing requires the exact project counts and hashes all arrays/edges. Conversion aborts on unknown cells, variable vertices under the fixed-vertex contract, branch-length mismatch, or excessive rejection fraction. Splitting prefers independent run groups. The training-split audit measures response support and proposes a finite cap. **freeze-config** binds geometry, dataset, split, audit, and cap by hash. Training rejects mismatched frozen provenance.

A schema-equivalent ROOT fixture was supplied. Binary inspection confirms the required branch strings, but the build environment lacked Uproot/Awkward and could not install them from its offline index. Numeric fixture and production inspection therefore remain mandatory CLI gates rather than inferred facts.

10 Evaluation and predeclared decisions

10.1 Structural gates

Every generated event must have finite nonnegative energies, no energy outside selected valid support, requested/realized count agreement, layer and event closure within 2×10^{-5} , and no response-cap violation.

10.2 Distributional evaluation

Required global and conditional diagnostics include total response, zero rate, ECAL/HCAL sharing, longitudinal means and covariance, first/last active layer, late energy, counts, positive-hit spectra, centroids, widths, containment radii, top-cell fractions, connectedness, and neighbor correlations. Both high-level and geometry-aware low-level classifier two-sample tests are required, with a truth-half statistical floor. Diversity, coverage, nearest-neighbor memorization, repeated identical- p^μ ensembles, and frozen downstream reconstruction closure are mandatory.

10.3 Primary gates

Before final-test exposure, the project freezes at least:

- absolute global mean-response bias $\leq 1\%$;
- absolute relative resolution difference $\leq 5\%$;
- no energy-bin mean bias above 2%;
- no energy-bin resolution difference above 10%;
- high-level C2ST AUC ≤ 0.60 ;
- low-level geometry-aware C2ST AUC ≤ 0.65 ;
- measured end-to-end speedup over the declared Geant4 reference $\geq 50\times$;
- no structural failure and satisfaction of the matched baseline rule across three seeds.

These are project engineering thresholds, not universal detector-simulation standards. Uncertainties and all component metrics remain visible.

11 Baselines and external context

The required internal ladder contains a conditional empirical/template model (B0), a matched non-graph CFM (B1), a single-stage graph flow (S1), a causal/serial ablation (A1), and a sparse point-cloud model (P1). P1 is required because recent post-reconstruction benchmarking indicates a favorable speed-accuracy balance for point-cloud calorimeter surrogates [7, 8].

Relevant external work includes ALICE ZDC generators [9, 10, 11, 12], inductive/layerwise flows [13, 14], and irregular-geometry graph generation [15]. Public methods must first be reproduced on their native detector representation. Their adaptation to this 6,790-channel layered ZDC is a new experiment, not a direct reproduction. For example, the public flow-matching ZDC work reports excellent native ALICE latency, but the channel count, topology, scoring, and hardware protocol differ [11].

All adapted comparisons freeze data, split, target, geometry, condition, optimization budget, seeds, solver evaluations, metrics, and timing scope. Parameter count alone is not a fairness criterion.

12 Software QA and verified execution

The active source compiles and the bundled regression suite currently contains 18 passing tests. The tests cover energy convention and mass-shell handling, target-mode contracts, sparse exact decoding, preselected support, geometry/data artifacts, split-group integrity, model structural invariants, flow-time dependence, configuration validation, and loss-weight calibration. This is a minimum executable regression suite rather than a claim of exhaustive coverage. Production ROOT I/O, full CLI orchestration, long-running GPU training, and large-scale evaluator behavior remain mandatory production QA.

An end-to-end synthetic CLI run executed:

1. synthetic geometry/data generation;
2. deterministic split;
3. training-split audit;
4. configuration freeze by hash;
5. one-epoch joint training;
6. checkpoint reload;
7. sample generation;
8. structural QA;
9. frozen evaluation and gate execution;
10. model-only timing.

The structural report contained no nonfinite, negative, outside-support, count-mismatch, closure, dust, or cap violations; its reported event/layer closure residual was zero at the configured report precision and remained within the declared 2×10^{-5} GeV tolerance. The deliberately tiny one-epoch model failed the physics gates, demonstrating that the evaluator does not convert software execution into a false fidelity pass. This is software evidence only. Synthetic events do not validate the production ROOT adapter at scale, the actual detector mapping, or Geant4 fidelity.

13 Residual risks and falsification paths

The remaining risks are empirical rather than hidden in the active mathematical contract:

- production ROOT units, primary selection, fixed vertex, sentinel, and ganging may differ;
- the strict 6,790-channel map may not be recovered from production hits without an external detector manifest;
- full graph training may exceed T4 memory or latency targets;
- the response mixture may miss tails or endpoint atoms;
- p4 alone is insufficient if the simulation varies vertex or beam conditions;
- a non-graph or point-cloud baseline may dominate the proposed hierarchy;
- exact support can preserve sparsity while count or spatial morphology remains wrong.

Each risk has a direct measurement or ablation. A negative result under the frozen protocol is a valid outcome and must not be obscured by post-test metric or target changes.

14 Auditor checklist

Before any physics claim, an independent auditor should verify:

1. source, environment, ROOT, geometry, data, split, audit, config, and checkpoint hashes;
2. exact energy/unit/target semantics and training-only cap estimation;
3. strict detector counts and edge/connectivity report;
4. no train-test leakage or post-test adaptation;
5. component and free-running metrics across three seeds;
6. structural invariants on large generated samples;
7. binned high/low-level, diversity, memorization, and reconstruction studies;
8. matched baseline fairness and native external reproductions;
9. decomposed batch-1 and throughput timing against measured Geant4;
10. claim language consistent with the actual passed gates.

15 Conclusion

The v2.2 revision converts the earlier strong but internally incomplete proposal into a coherent CLI-ready experiment. Its key design commitments are conservative: correct incident-energy semantics, mutually exclusive target modes, arbitrary nonnegative longitudinal profiles, no unidentified slack or latent, one exact support mechanism, continuous flow only where the target is continuous, strict provenance, and predeclared failure gates. These choices make the experiment auditable and structurally valid. They do not predetermine whether the model will be an accurate or fast ZDC surrogate. That question is now delegated to a frozen, reproducible empirical program on the actual 765k-event Geant4 corpus.

A Tensor and artifact contract

Object	Shape/type	Contract
p4_total_gev	float $[B, 4]$	$[E_{\text{tot}}, p_x, p_y, p_z]$ in GeV; neutron mass shell checked in float64.
kinetic_energy_gev	float $[B]$	$E_{\text{tot}} - m_n$; used for selection, binning, and response scaling.
cell_energy_gev	float $[B, 6790]$	Frozen readout order; finite and nonnegative.
layer_index	int $[6790]$	Values $0, \dots, 64$ with strict project counts.
node_features	float $[6790, F]$	Standardized position, layer, and subdetector metadata only.
edge_index	int $[2, E]$	Deterministic physical graph with in-range indices.
edge_features	float $[E, 7]$	$\Delta x, \Delta y, \Delta z, d$ and edge-type indicators.
visible	float $[B, 1]$	One iff modeled target response is positive.
total_response_gev	float $[B, 1]$	Sum of modeled cell response.
first_positive_layer	int $[B]$	-1 for zero response; otherwise first positive layer.
active_layers	bool/float $[B, 65]$	Exactly those layers assigned positive budget.
layer_energy_gev	float $[B, 65]$	Nonnegative and summing to total response.
requested_counts	int $[B, 65]$	Geometry/budget-feasible categorical samples.
support_mask	bool $[B, 6790]$	Exactly requested count per layer.
share_state	float $[B, 6790]$	Continuous flow state; zeroed outside support every step.
geometry_manifest.json	JSON	Counts, edge contract, and geometry hash.
dataset_manifest.json	JSON	Source hashes, target semantics, geometry hash, rejects, shards.
splits.json	JSON	Grouping rule, seed, group count, and immutable indices.
train_data_audit.json	JSON	Training-only support, cap, vertex, duplicate, and shell diagnostics.
best.pt/last.pt	PyTorch	Model, optimizer, scheduler, scaler, hashes, stage, epoch, seed.

B Experiment matrix

Table 3: Minimum experiment matrix. All rows use three predeclared training seeds.

ID	Training support	Model	Purpose
G1-full	0–300 GeV	full v2.2 graph hierarchy	primary broad-support experiment
G1-primary	50–250 GeV	matched full v2.2	tests whether outer support helps the primary domain
B0	matched	empirical/template	minimum competent non-neural reference
B1	matched	non-graph CFM hierarchy	tests value of geometry graph
S1	matched	single-stage graph flow	tests value of explicit hierarchy
A1	matched	causal/serial ablation	tests depth-causal restriction and latency
P1	matched	sparse point cloud	tests alternative sparse representation
Native-ZDC	native public data	public ZDC methods	reproduction before detector adaptation

Every matched row freezes the target, condition, geometry, split, optimization budget, solver evaluations, seeds, and metric implementation. The comparison reports parameter count, peak memory, training time, batch-1 latency, throughput, and the full physics vector.

C Failure-mode matrix

Observed failure	Immediate diagnosis	Disposition
Extreme or capped response	energy convention, cap, mixture support, units	stop; re-audit training data and re-freeze
Dense low-energy dust	support-mask bypass or dense decoder	hard structural failure
Correct total, wrong counts	count calibration and generated-budget feasibility	repair count stage; do not retune total loss
Correct counts, wrong morphology	support ranking, graph, geometry, low-level C2ST	inspect geometry and support stage
Late HCAL suppression	start/activity/profile conditional distributions	compare teacher-forced and free-running stages
Good stage losses, bad cascade	generated-condition exposure bias	staged free-running fine-tuning
Near-perfect C2ST	globally separable samples	reject; use classifier feature importance to localize
Empty split	insufficient independent run groups	repair provenance or disclose event-hash fallback
Unknown cell ID	sentinel/ganging/geometry mismatch	abort conversion; new geometry experiment if corrected
GPU memory failure	graph batch/hidden/edges too large	reduce batch first; profile before architecture change
Fast model, failed reconstruction	low-level mismatch hidden by aggregate metrics	reject primary physics claim
Good fidelity, insufficient speed	solver/backbone/decoder bottleneck	optimize only under frozen fidelity gates

D Audit-to-code traceability

Audit concern	Active implementation	Verification
Kinetic/total ambiguity	<code>contracts.py</code> , <code>data/root_io.py</code> , <code>data/convert.py</code>	shell/energy unit tests
Mixed threshold accounting	<code>TargetMode</code> , converter, truth hierarchy, decoder	raw and threshold decoder tests
Shared latent ignored	no latent argument/state in active model	source/API tests
Reserve non-identifiability	no reserve variable; direct closure	profile/system tests
Double support randomness	<code>SupportScoreField</code> + one support sample	seed/exact-support tests
Singular support CFM	<code>ShareFlowField</code> only on selected continuous shares	flow mask/time tests
Count infeasibility	dynamic categorical logit mask	count/decoder tests
Geometry ambiguity	scan/freeze/hash and strict counts	hash mutation/count tests
Leakage-prone split	group hash with empty-split refusal	split tests and manifest
Unfrozen response cap	training-split audit + config freeze	synthetic audit/freeze smoke
Unfrozen RNG/optimizer	seed bundle and hashed resolved config/checkpoint	reproducibility/checkpoint tests
No success gates	gate YAML and evaluator	gate parser/application tests
Bibliography errors	rebuilt primary bibliography	metadata audit log

E Production command transcript template

An execution record should preserve the following literal command sequence and outputs:

```
python -m pip install --no-build-isolation -e '[root,eval,dev]'
python -m compileall -q src tests vertex
pytest -q
cbssc-zdc inspect-root FILE --schema configs/schema_production_zdc.yaml
cbssc-zdc scan-geometry FILES --schema ... --output artifacts/geometry
cbssc-zdc convert FILES --schema ... --geometry ... --output artifacts/data \
--target-mode raw_deposit --min-kinetic-gev 0 --max-kinetic-gev 300
cbssc-zdc split --manifest artifacts/data/dataset_manifest.json \
--output artifacts/splits.json --group-by source_group --seed 20260723
cbssc-zdc audit-dataset --manifest artifacts/data/dataset_manifest.json \
--splits artifacts/splits.json --split train --kinetic-range 0 300 \
--output artifacts/train_data_audit.json
cbssc-zdc freeze-config --template configs/templates/train_full_0_300_raw.yaml \
--audit artifacts/train_data_audit.json --geometry artifacts/geometry \
--manifest artifacts/data/dataset_manifest.json --splits artifacts/splits.json \
--output configs/frozen_full_0_300.yaml
cbssc-zdc train --config configs/frozen_full_0_300.yaml
cbssc-zdc qa --geometry artifacts/geometry --checkpoint runs/.../best.pt --device cuda
cbssc-zdc evaluate --checkpoint runs/.../best.pt --geometry artifacts/geometry \
--manifest artifacts/data/dataset_manifest.json --splits artifacts/splits.json \
--split test --gates configs/gates_primary.yaml --output runs/.../test.json \
--device cuda
cbssc-zdc benchmark --checkpoint runs/.../best.pt --geometry artifacts/geometry \
--device cuda --batch-size 1
```

F Evidence boundary for the supplied fixture

The supplied ROOT fixture is explicitly not drawn from the production event population. Its role is limited to branch/schema compatibility. The bundle records its SHA-256 and confirms the expected branch strings by binary inspection. Numerical event, unit, geometry, and distribution checks remain unresolved until the production environment installs Uproot/Awkward and executes the CLI. This limitation prevents a false claim that the 6,790-channel mapping or response distribution was recovered from the fixture.

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